GFS - GlobalFoundries | Full investment artifact

Research update: October 14, 2026 | **Framework:** 8.0 | **Financial model:** Japan-SPV net OP and conditional government-ownership risk; original products and interposer paths retained **Status:** NOT HELD in supplied original | **Classification:** BOTTLENECK CANDIDATE - differentiated photonics/SiGe within a diversified foundry **Artifact Base:** 40/60 = **67% - Watchlist** | **Portfolio Excel Base:** 67% (current October 7 owner setting retained; separate from artifact assessment) **Active modeled ceiling:** Q4 2027 **USD 76.360690**, pricing Q4 2027-Q3 2028 at retained 20x non-IFRS OP, 562M target-quarter base shares and a separate expected ownership factor 0.976. Final SPV agreements, fee accounting and actual government share-transfer terms remain unresolved.

The forward opportunity is larger than the present CID revenue share suggests. Silicon photonics, SiGe and power can increase both market access and content per system. The full model carries that expansion into product GP while retaining the slower-growth foundry base, rising R&D and equity compensation allowance.

1. Supercycle, supply chain and durable advantage

The investment case is a mix shift toward differentiated optical, power and embedded technologies within a broad foundry. Photonics and SiGe are complementary, but neither implies that the entire company earns optical margins. New customer agreements support demand visibility; tool delivery, qualification and profitable capacity utilization determine the conversion to earnings.

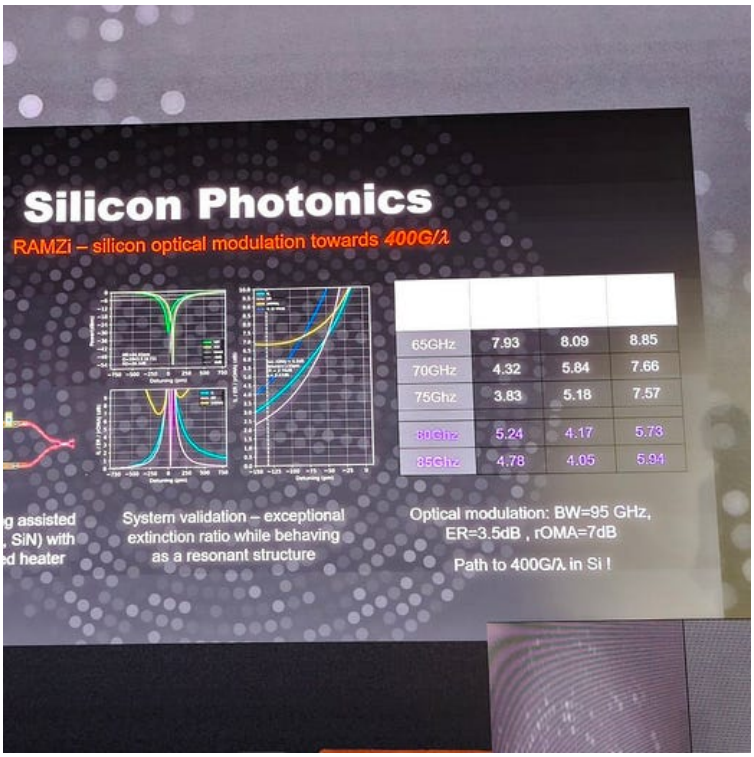
DB1 photonics and DB10 inference: optical bandwidth and electrical power demand → customer chip designs → qualified SiPho, SiGe and BCD processes → wafer fabrication and selected integration services. GF’s process design kits, yield history and diversified footprint create qualification advantages. It is not a monopoly, and customers can use different foundry routes across products.

Business / curve	Stage	Evidence and next inflection
Pluggable SiPho and SiGe	Commercial ramp;10-50% qualitative	Current demand and signed capacity agreements support added qualified production.
NPO/CPO SCALE	Pre-ramp to early ramp	Tape-outs must pass qualification before recognized volume revenue.
Data-center power and physical AI	Early ramp	New processes and design partnerships broaden content; revenue timing varies.
Mobile/auto and legacy markets	Mixed mature/recovery	Not every end market shares the optical growth rate.

Stage percentages are qualitative framework bands relative to a possible mature opportunity, not measured market penetration. A current capacity constraint can coexist with a larger long-term TAM; qualification and economic capture decide how much reaches this company.

Engineering: compare an optical link, not a technology label

The February 12 Irrational Analysis memo makes a strong negative comparison between GlobalFoundries and Tower. Its useful engineering questions concern modulator bandwidth, optical loss, drive power, packaging and the qualified customer's complete link. Its language about an inferior process and inferred customer concentration goes beyond the evidence reproduced. A company can have successful SiGe, coherent-optics or silicon-photonics products without winning every CPO architecture. Conversely, a research demonstration does not establish production yield or commercial competitiveness. [E-GFS1]



Engineering exhibit tower-I12. Saved source: (Market Memo) Tower+GloFo Earnings, HBM4 Qualification Noise, 2026-02-12. Attribution and scope are discussed below.

RAMZI technical-slide photograph reproduced by the author. It presents device-level performance and a path toward 400G per wavelength. It is not a production-yield distribution, a customer allocation or a demonstrated 400G commercial module.

RAMZI combines interferometric and resonant elements. A fair comparison needs the same wavelength, electrical drive, optical input/output boundary, temperature range and receiver-error criterion. More bandwidth is useful only if the complete link retains adequate optical modulation amplitude, extinction ratio and noise margin at acceptable power and cost. Thermal control, packaging loss and manufacturing variation can alter the result. A small-signal bandwidth number alone cannot price a commercial wafer or determine the winning foundry.

The article assigns 90% confidence to an unnamed CPO win being Cisco and estimates that Cisco/Acacia represents 80-90% of GF's SiPho business. Those are source-authored estimates, not disclosed customer shares. The current model does not adopt those probabilities as measured allocations, nor infer that acquiring AMF or Infinilink proves failure of GF's existing platform. Process portfolio expansion, IP and engineering capability are alternative explanations; customer-level revenue and qualification evidence would distinguish them. [E-GFS1]

Current probability and cost treatment. Retain **70% conversion confidence** for the existing SCALE NPO/CPO programme, **85% execution confidence** for the Marvell SiGe agreement and **80% timing confidence** for MPS production. These are the existing ledger's confidence labels, not new haircuts or additive order values. The model already has SiPho sales of **$585M in 2027 and $900M in 2028**, with explicit product GMs of 52% and 55%; manufacturing depreciation stays in costs and R&D/opex rises in dollars. The February claims supply no newly identified lost customer, failed qualified process or price concession relative to the newer September evidence. Their incremental sales, COGS, opex and OP are zero; they do not justify a whole-company haircut or a transfer of revenue to Tower.

The engineering risk is nevertheless concrete. A design that needs more drive/laser power, more thermal correction, larger area or additional test/rework can have a lower accepted yield and higher cost per billed unit. The appropriate negative branch would isolate the exposed customer/process and recognition quarter, retain unavoidable fab and R&D costs, and net any substitution to another GF product. The appropriate positive branch would require a defensible increase in accepted units or net content after cannibalization. Neither a photographed eye diagram nor broad AI optical demand sizes that incremental invoice.

Newer Marvell and MPS agreements remain primary company evidence and are already inside other CID and the appropriate residual end markets. The model's $1.08B SiPho exit-2028 annualized rate is distinct from its $900M full-year sales. The engineering claims add no separate economics. After the separate interposer and October 14

SPV/ownership treatments, the **Q4 2027 price target is $76.360690**, retaining the existing 20x non-IFRS OP method with 562M base shares and the separate 0.976 expected ownership factor. It does not change the current Excel Base of 67 or the separate analytical Base of 67.

Engineering sources, reviewed October 6, 2026: E-GFS1: Irrational Analysis, February 12, including the saved GF call excerpts and RAMZI photographs. G4: Marvell capacity partnership, September 17. G5: MPS partnership, September 9. The original technical-slide paper was not independently identified from the saved photograph, so its plotted results remain an attributed exhibit rather than a reproduced experiment.

GaN manufacturing: follow the wafer, then qualification

The May 17 Irrational Analysis article identifies GlobalFoundries as a manufacturing route behind Navitas's integrated GaN technology. Newer primary evidence provides a much clearer milestone: the September 1 joint announcement schedules first Gen5 GaNFast **wafers in September**, internal samples in October and strategic customer samples before year-end. The process uses GF's US 200mm GaN-on-silicon manufacturing platform. This verifies a named programme and a staged manufacturing route; it does not say that qualified AI power systems are already shipping in volume. [E-GFS2, E-GFS3]

The value split matters. Navitas sells the power devices and integrated functionality; GF earns the applicable foundry invoice. End-system watts, a transistor's maximum heat dissipation and Navitas device revenue are not interchangeable with GF wafer sales. Better integration can reduce customer board area or protection effort, but GF must convert process qualification into good wafers, customer acceptance and profitable utilization. Startup scrap, added process steps, test and depreciation can offset a richer selling price.

Probability and bounded treatment. The announced schedule is a verified statement, while execution remains uncertain. Assign **75% analyst confidence** to strategic customer sampling by the end of 2026. For financial recognition, use a bounded diagnostic of **0-1% of the existing $1,150M CY2027 other-CID forecast**, or **$0-11.5M** of possible Navitas-related foundry revenue within that category. The zero lower bound reflects sample-to-volume timing; the upper bound is a deliberately small attribution proxy, not a disclosed wafer commitment. Probability weighting gives $0-8.625M before the **100% overlap adjustment** against the current power/communications growth category.

At the category's existing 40% product-GM assumption, the conditional range implies up to **$6.9M of manufacturing cost and $4.6M of GP**, before allocated development/customer-support expense. Early programme margin may be lower; those costs cannot be omitted to turn a sampling announcement into pure profit. The current model's other-CID trajectory already includes data-center power alongside SiGe and communications, and the consolidated opex path rises from $1,180M in 2027 to $1,380M in 2028. Incremental revenue, COGS, opex and OP remain **zero** after overlap. The new evidence identifies a plausible constituent of that demanding growth path rather than enlarging it without a wafer/price bridge.

This treatment is not a requirement to wait for official revenue guidance. A credible customer wafer schedule, qualified usable capacity, billable pricing or newly identified supply restriction could change the active forecast before reported revenue. The immediate checks are October internal samples, year-end customer samples and conversion into accepted production wafers. Neither the announcement nor the older article establishes exclusive supply, profitable high-volume yield or a separately additive contract value.

Q4 2027 is $76.360690 including the separate interposer and SPV/ownership treatments, using the retained 20x forward non-IFRS OP method, 562M base shares and the separate 0.976 expected ownership factor, current Excel Base67 and analytical Base67. Existing SCALE70%, Marvell85% and MPS80% confidence labels are not applied again to the same sales.

Sources: E-GFS2: Irrational Analysis, May 17, 2026, including its integrated-driver and GaN manufacturing discussion; E-GFS3: GF/Navitas Gen5 US manufacturing milestone, September 1, 2026. The source timetable distinguishes first wafers, internal samples and customer samples; later volume recognition remains a model assumption.

October 11: TSMC interposer agreement

TSMC interposers: a new invoice, with a later factory ramp

GlobalFoundries announced on October 8 a $2 billion manufacturing agreement with TSMC, initially lasting five years. GF plans additional Malta capacity for silicon interposers supporting CoWoS, including embedded deep-trench capacitor components, with volume production expected to begin ramping in the first half of 2028. The disclosed agreement is a verified event; its timing and profitable delivery remain forward-looking. This manufacturing service is distinct from selling a photonic wafer or an entire AI package.

The existing model already includes selective 2027 pricing, the Marvell SiGe relationship, MPS and a substantial SiPho/other-CID expansion. The TSMC interposer contract was absent from that earlier source register, but generic other-CID growth can overlap the new business. We therefore reserve 50% of the conditional programme revenue against that existing growth allowance, leaving SiPho forecasts unchanged. This is an explicit model overlap assumption, not a disclosed customer allocation.

The programme's existence is established. For execution, the model uses 85% probability on a bounded accepted-sales schedule: $200 million gross programme sales in 2028, within a $100-300 million conditional range, followed by $400 million in each of 2029 and 2030. The 2028 profile is $0/$25/$75/$100 million, starting in Q2 within the issuer's first-half ramp window. Subsequent quarters are $75/$100/$100/$125 million in 2029 and $100 million each in 2030. Those annual and quarterly allocations are analytical assumptions, not contracted minimum purchases. The gross model schedule totals $1 billion through 2030; the undistributed remainder of the headline agreement is not accelerated into the current forecast or treated as perpetual sales.

After the 85% execution weight and 50% overlap reserve, additional sales are $85 million in 2028 and $170 million in each of 2029 and 2030. The programme gross-margin assumptions are 30% at ramp and 35% subsequently, including manufacturing depreciation, process yield, scrap and startup factory costs. They are below the existing other-CID margins of 43%/45%/46%. A further 5% of incremental revenue funds engineering and customer support, producing $21.25 million incremental operating profit in 2028 and $51 million in each of 2029 and 2030.

Qualification must be paid before successful sales. We use a $10 million conditional engineering budget, 0.5% of the disclosed agreement amount, with 50% already covered by existing operating expenses. The remaining $5 million is expensed $2.5 million in each of Q3 and Q4 2027. Those expected incurred costs are not discounted by the probability that production ultimately succeeds. Manufacturing startup costs remain in the product-margin assumption, avoiding a second charge for the same factory activity.

The active Q4 2027 window captures the Q4 2027 qualification cost and the first three quarters of 2028, not the full 2028 contract contribution. Its incremental forward operating profit is $8.125 million. The retained method remains 20 times non-IFRS operating profit, including manufacturing depreciation but excluding recurring SBC, divided by the target quarter's existing 562 million diluted-share proxy. The interposer branch itself changes no cash/debt treatment, multiple, underlying share schedule or score. The separate October 14 ownership uncertainty is handled below. Excel Base67 and analytical Base67 are retained; old artifact wording describing a protected Excel Base63 predates the owner's October7 override removal.

The primary announcement does not disclose minimum purchase enforceability, annual orders, payment composition, capex, usable wafer capacity, customer advances, subsidy terms or a product-margin bridge. The numerical case uses bounded proxies for those incomplete operating details rather than assuming that all $2 billion is near-term, pure-margin revenue. Material dedicated capex, poor yield, qualification delays or displacement of more existing production than the overlap reserve would reduce the outcome. The next useful evidence is Malta tool installation and customer qualification, the accepted shipment schedule, and the relationship between manufacturing invoices, funding and depreciation.

Source: GlobalFoundries, October 8, 2026, https://gf.com/news-and-events/news/globalfoundries-reaches-agreement-to-establish-us-based-supply-of-silicon-interposers-for-advanced-ai-packaging/

Quarter	Conditional sales	Net added sales	Net added GP	Extra opex	Added OP
Q3 2027	0.000	0.000	0.000	2.500	-2.500
Q4 2027	0.000	0.000	0.000	2.500	-2.500

Quarter	Conditional sales	Net added sales	Net added GP	Extra opex	Added OP
Q1 2028	0.000	0.000	0.000	0.000	0.000
Q2 2028	25.000	10.625	3.188	0.531	2.656
Q3 2028	75.000	31.875	9.562	1.594	7.969
Q4 2028	100.000	42.500	12.750	2.125	10.625
Q1 2029	75.000	31.875	11.156	1.594	9.562
Q2 2029	100.000	42.500	14.875	2.125	12.750
Q3 2029	100.000	42.500	14.875	2.125	12.750
Q4 2029	125.000	53.125	18.594	2.656	15.938
Q1 2030	100.000	42.500	14.875	2.125	12.750
Q2 2030	100.000	42.500	14.875	2.125	12.750
Q3 2030	100.000	42.500	14.875	2.125	12.750
Q4 2030	100.000	42.500	14.875	2.125	12.750

The $2B headline is not recognized as revenue at signing. The modeled 2028-30 gross sales sum $1B before probability and overlap. The remainder is not automatically booked, because the initial term dates and payment mix are undisclosed. A capital-budget diagnostic of 20-40% of the headline value is $400-800M before any customer funding, reuse or subsidy. This is a financing stress range, not a new cash-flow forecast. Manufacturing depreciation and startup loss are already charged through the branch gross margins. No debt or share change is asserted without financing evidence.

Japan-funded Malta expansion: fees before first wafers

GF's October 14 release describes up to USD 21 billion of Japanese investment for two new Malta fabs. The initial approximately USD 10 billion phase is expected to come online in 2029; a second approximately USD 10 billion phase remains subject to approval. Each phase's approximately 900,000 300mm-equivalent wafers per year is eventual capacity at full operation, not first-year good-unit shipments. The first site is permitted and shovel-ready. This is a financing, technology and operating arrangement, not a new earnings release. [G9, G11]

The investor deck supplies the company-level capture mechanism. A U.S.- and Japan-owned special-purpose vehicle funds and owns the fab infrastructure, tools and operating expenses. GF supplies technology, construction and operating management, and earns technology fees, construction/operating management fees, EBITDA sharing and shared-services reimbursement. The SPV's wafer invoices and the investment headline are not GF sales. GF's associated engineering, administration, labor and allocated shared-services costs must be netted against its fees; funded equipment is not a free cash balance for GF shareholders. [G10, PDF pages 10-12]

Technology fees begin in 2027 and run for at least 20 years. Phase-one-only management illustrations start at 30-50 basis points of annual non-IFRS gross-margin accretion, 100-200 basis points of operating-margin accretion and USD 0.10-0.15 diluted-EPS accretion in 2027. Long-term illustrations are 100-200/200-300 basis points and USD 0.50-0.75 respectively, with about USD 3 billion cumulative adjusted free cash flow over the first ten years. They assume January 1, 2027 commencement and remain subject to transaction closing, definitive contracts and final accounting. EPS is an after-tax cross-check, not a valuation input. The primary deck gives no fee split, quarterly revenue/expense classification, margin denominator or share assumptions. [G10]

Bounded operating case. ASTRA assigns 60% execution probability and a 25% overlap reserve against existing technology-services growth and cost leverage. The specific SPV arrangement was absent from the earlier source register; the reserve recognizes that current services, mix and utilization paths already assume substantial improvement. Conditional net operating contribution is each existing quarter's reference revenue multiplied by 1.5% in 2027 and 2028, 2.0% in 2029 and 2.5% in 2030. The 2027 midpoint sits within the issuer's 1-2% operating-margin illustration; the 2028 carry-forward and later stepped rates are bounded analyst assumptions, not management annual guidance. Quarterly pacing follows reference revenue, an analytical proxy rather than an issuer fee schedule.

This net-OP branch is **already after program costs**. It represents technology/construction/operating fees and eligible reimbursements/EBITDA sharing less associated engineering, administration, labor and allocated shared-services costs. Gross fee revenue and its COGS/opex presentation remain unallocated. It is not product gross profit, pure-profit revenue or a claim that costs are zero. Updated OP equals existing product GP minus existing opex plus the separate net contribution. Because reference sales exclude unallocated Japan fees, OP/reference revenue is explicitly a proxy rather than reported consolidated operating margin.

The conditional 2027 OP range is USD 82.950-165.900 million on the existing USD 8,295 million revenue reference, with USD 124.425 million at the 1.5% midpoint. Applying 60% execution and retaining 75% after overlap gives **USD 55.991250 million** expected annual contribution. The active Q4 2027-Q3 2028 window contains **USD 62.724375 million** expected net contribution. No second-fab contribution, grant windfall, capacity-times-ASP revenue or duplicate TSMC interposer economics is added.

Separate conditional government-equity risk. The release and SEC exhibit say Commerce will receive approximately 4% equity in GF, but establish neither new issuance versus an existing-holder transfer nor pre/post basis, share count, price, consideration, timing or warrants. The operating arrangement and the equity agreement are separate. The model assigns a 60% new-common-issuance scenario and 40% existing-holder transfer/no-incremental-share scenario. The issuer being a party to the agreement supports a plausible new-issue case, while the missing terms leave substantial alternative-structure risk. These are analyst judgments, not reported probabilities or a confirmed issue.

If Commerce received 4% of post-issue common equity through new issuance, state shares equal base shares divided by 0.96. In the transfer/no-issue state, shares remain unchanged. From target-quarter Q1 2027, the expected per-share factor is 40% x 1 + 60% x 0.96 = **0.976**. January 1, 2027 is the earliest assumed issuance date, not a confirmed date. The underlying 556/557/562/568/574/580 million share path remains visible; its SBC/acquisition allowance is separate from the government risk. Q4 2027 shares would be 585.416667 million in the full-new-issue state versus 562 million in the transfer state.

Execution and new issuance are assumed independent; actual dependence is unknown. Joint weights are 16% neither, 24% issuance without execution, 24% execution without issuance and 36% both. The operating-profit method is identical in each state. The weighted target equals the expected-OP calculation times the ownership factor. Signed agreements or funding conditions linking the outcomes would replace this simplifying assumption.

The separate USD 955 million CHIPS amendment reallocates an existing award toward SiPho, SiGe and advanced packaging at Malta/Burlington. It is not an incremental OP grant. Existing SiPho, Marvell, MPS and interposer forecasts already carry capacity/growth assumptions; the older USD 375 million quantum and USD 300 million photonics references are not added again. Remaining gaps include final fee recognition, direct/shared cost allocation, funded-capex responsibilities, closing dates, actual share-transfer mechanics and phase-two approval. Original PDF bytes and slide pixels were unavailable; full issuer/SEC release and deck text were recovered through supported web reads. No inaccessible visual detail is represented as reviewed.

Dated model comparison, October 14: the same-quarter Q4 2027 ceiling moves from USD 76.00622775800711 to **USD 76.36069003558720**, or **+0.466359518%**, after net SPV contribution and conditional ownership risk. The multiple, original operating paths, existing TSMC branch and Excel/analytical Base 67 are retained. The fee uplift before ownership risk is partly offset by the conditional dilution allowance; these are separate mechanisms.

2. Forward TAM and the three-phase expansion lens

Phase	Engineering opportunity	Demand-to-company bridge
Phase 1;100% thematic	Current optical links and differentiated foundry work	More links, higher lane speeds and new designs expand the served wallet. Wafer content depends on die area, integration and yield rather than a fixed fraction of optical-module ASP.
Phase 2;50% thematic	NPO/CPO, SiGe and data-center power	SiPho target framing extends from >1B exit 2028 toward 2B 2030 run rate. The model uses 900M/1,950M actual annual sales equivalents and 1.08B/2.16B exit annualized values. Marvell and MPS agreements support availability/capture without disclosed additive contract values.
Phase 3;20% thematic	Quantum, physical AI, embedded compute and wearables	New IP and process combinations can create design access before wafer revenue. Government R&D funding is milestone-based support, not immediate commercial revenue; services already sit inside reported end markets.

The 100% / 50% / 20% thematic weights guide thesis assessment. They are not blanket discounts to reported sales, contracts or all forecasts. Programme probabilities below govern only separately identified optionality. Current reported sales calibrate the starting point; they do not cap the future market.

3. Current earnings and management evidence

Q2 2026 revenue was USD 1,786M. Non-IFRS GP/OP were 534M/298M, versus IFRS 505M/174M. Q3 guidance midpoint is revenue 1,885M, non-IFRS GM 30.5% and opex 260M, implying OP 314.925M. The guided fully diluted count is 556M, replacing the old 548.42M denominator.

Reviewed the latest August 5 earnings release, substantive Q2 call passages and September 8 Goldman Sachs discussion, with a Q1 transcript cross-check. Complete sequential coverage of both earnings-call Q&As was not achieved. September 9 and September 17 agreements are incorporated. The September discussion reinforces supply/qualification constraints rather than a new quantified revenue guide.

Latest financial release, Q2 management remarks/Q&A passages, Q1 cross-check and the newer Goldman discussion were reviewed. Sequential coverage of both complete earnings calls is not claimed. Product margins, wafer-equivalent ASP and outer-year capture remain analytical assumptions, even where the direction is supported by signed partnerships.

4. Local evidence and calibration packet

The latest issuer release anchors Q2 actuals and the Q3 midpoint. The forward series starts in Q3 and uses estimated product allocations 85M SiPho+225M other CID+1,575M other=1,885M. Product GP 42.5+85.5+446.925=574.925M, exactly 30.5%; subtract 260M expenses for 314.925M OP. Initial product GM calibration is not evidence of disclosed segment costs. Q2 IFRS OP 174M versus adjusted 298M illustrates a 124M basis difference.

5. Summing thesis score

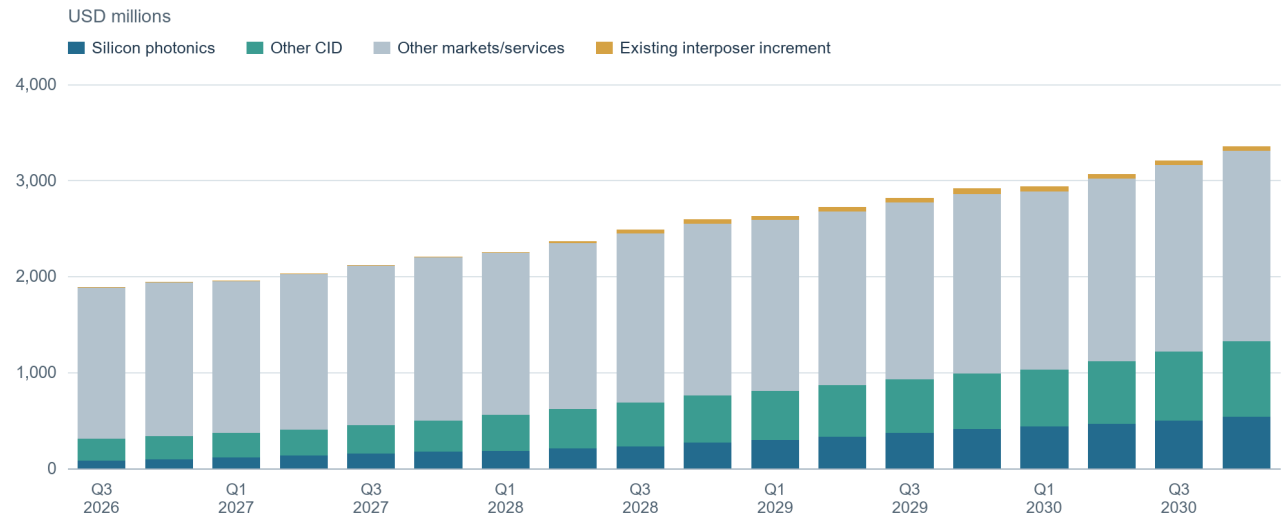
Category	Points / maximum	Evidence
Core Thesis	16/20	Verified optical capacity demand and customer commitment; competition and legacy exposure remain.
Cycle	7/10	Optical ramp offsets mobile weakness; capacity installation gates conversion.
TAM	8/10	NPO/CPO, data-center power and physical AI broaden available content.
Catalysts	5/5	Marvell capacity agreement, MPS production in 2027 and SCALE qualifications.
Macro	7/10	AI demand and diversified manufacturing benefit; handset and auto cycles still matter.
Alpha	2/5	Differentiation is increasingly visible and publicly promoted.
Risks	-5	Capex, customer concentration, ramp/yield, competition, recurring SBC and geopolitical risk.

Sum: 40/60 × 100 = 66.666667%, conventionally rounded to 67%. This advisory thesis score does not replace any portfolio Base override. The separate risk classes below total the displayed deduction; holding status and classification do not waive that deduction.

Financial views of the current model

USD millions except percentages and per-share values. Existing sales and product GP are reference series excluding unallocated Japan-SPV gross fees. Updated OP includes the separate net contribution after program costs. The original engineering exhibit remains; all financial charts below are regenerated from the exact staged model.

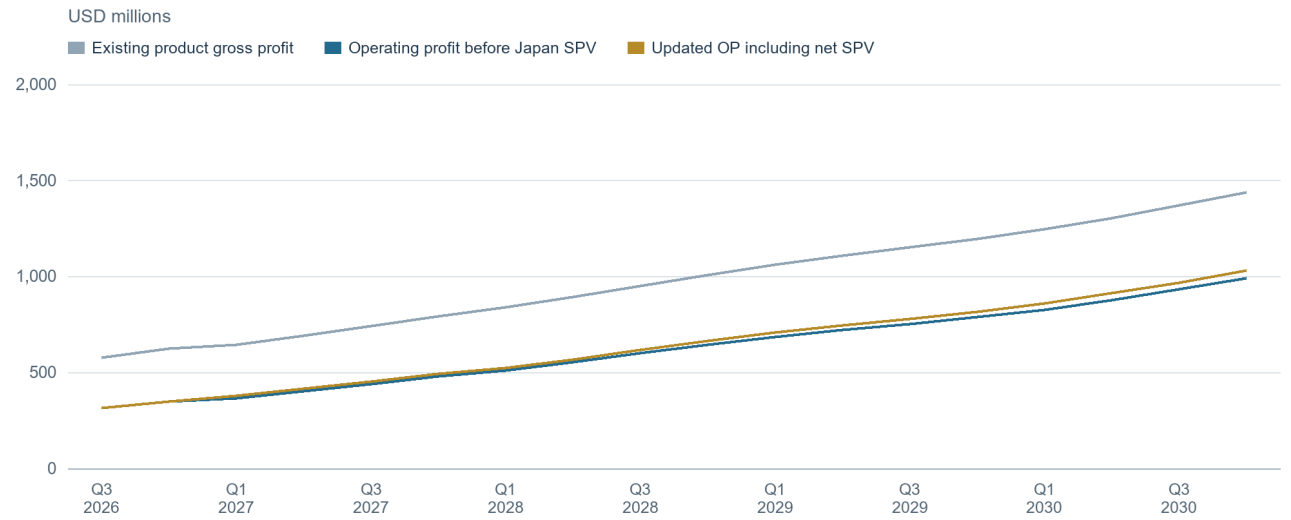
Revenue reference by existing business



Existing product revenue reference; Japan gross fees remain unallocated. This is not a complete future consolidated-sales presentation.

Quarter	SiPho	Other CID	Other markets	Interposer	Revenue ref.
Q3 2026	85.000	225.000	1,575.000	0.000	1,885.000
Q4 2026	100.000	240.000	1,600.000	0.000	1,940.000
Q1 2027	120.000	250.000	1,580.000	0.000	1,950.000
Q2 2027	135.000	275.000	1,620.000	0.000	2,030.000
Q3 2027	155.000	300.000	1,660.000	0.000	2,115.000
Q4 2027	175.000	325.000	1,700.000	0.000	2,200.000
Q1 2028	185.000	375.000	1,690.000	0.000	2,250.000
Q2 2028	210.000	410.000	1,730.000	10.625	2,360.625
Q3 2028	235.000	455.000	1,760.000	31.875	2,481.875
Q4 2028	270.000	490.000	1,790.000	42.500	2,592.500
Q1 2029	300.000	510.000	1,780.000	31.875	2,621.875
Q2 2029	335.000	535.000	1,810.000	42.500	2,722.500
Q3 2029	375.000	555.000	1,840.000	42.500	2,812.500
Q4 2029	415.000	575.000	1,870.000	53.125	2,913.125
Q1 2030	440.000	590.000	1,860.000	42.500	2,932.500
Q2 2030	470.000	650.000	1,900.000	42.500	3,062.500
Q3 2030	500.000	720.000	1,940.000	42.500	3,202.500
Q4 2030	540.000	790.000	1,980.000	42.500	3,352.500

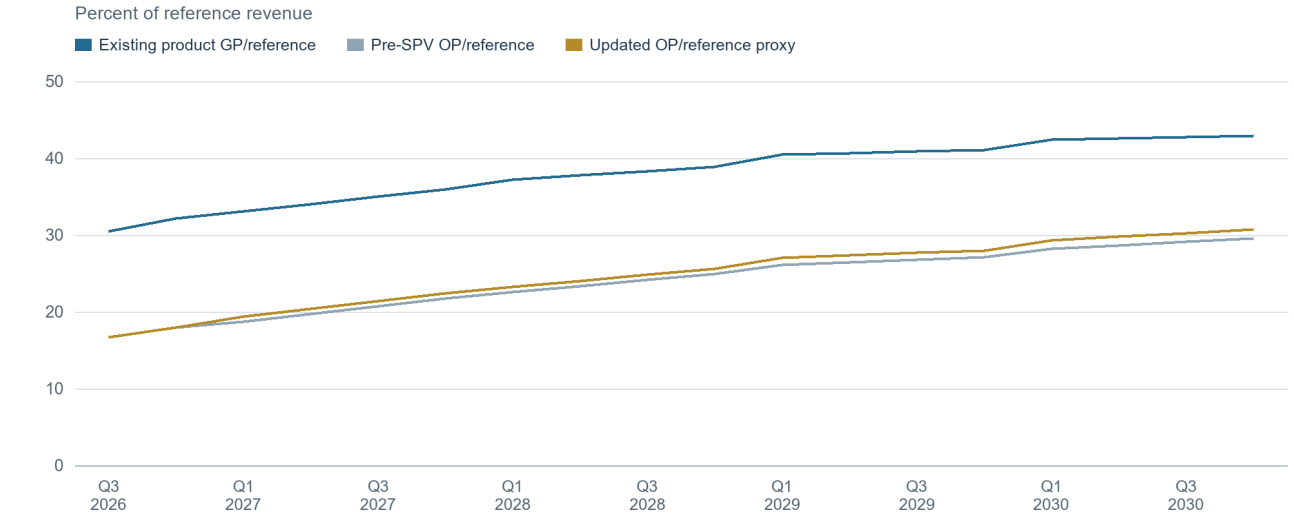
Product gross profit and operating-profit bridge



Net Japan-SPV operating contribution is separate from product GP and already includes program costs.

Quarter	Product GP	Pre-SPV OP	Net SPV OP	Updated OP
Q3 2026	574.925	314.925	0.000	314.925
Q4 2026	623.600	348.600	0.000	348.600
Q1 2027	644.300	364.300	13.163	377.462
Q2 2027	690.500	400.500	13.702	414.203
Q3 2027	740.100	437.600	14.276	451.876
Q4 2027	790.500	478.000	14.850	492.850
Q1 2028	837.600	507.600	15.188	522.788
Q2 2028	891.838	551.306	15.934	567.240
Q3 2028	950.462	598.869	16.753	615.621
Q4 2028	1,007.400	645.275	17.499	662.774
Q1 2029	1,061.356	684.762	23.597	708.359
Q2 2029	1,107.225	720.100	24.503	744.602
Q3 2029	1,149.975	752.850	25.312	778.162
Q4 2029	1,196.444	788.787	26.218	815.006
Q1 2030	1,243.875	826.750	32.991	859.741
Q2 2030	1,303.775	876.650	34.453	911.103
Q3 2030	1,368.275	931.150	36.028	967.178
Q4 2030	1,438.475	991.350	37.716	1,029.066

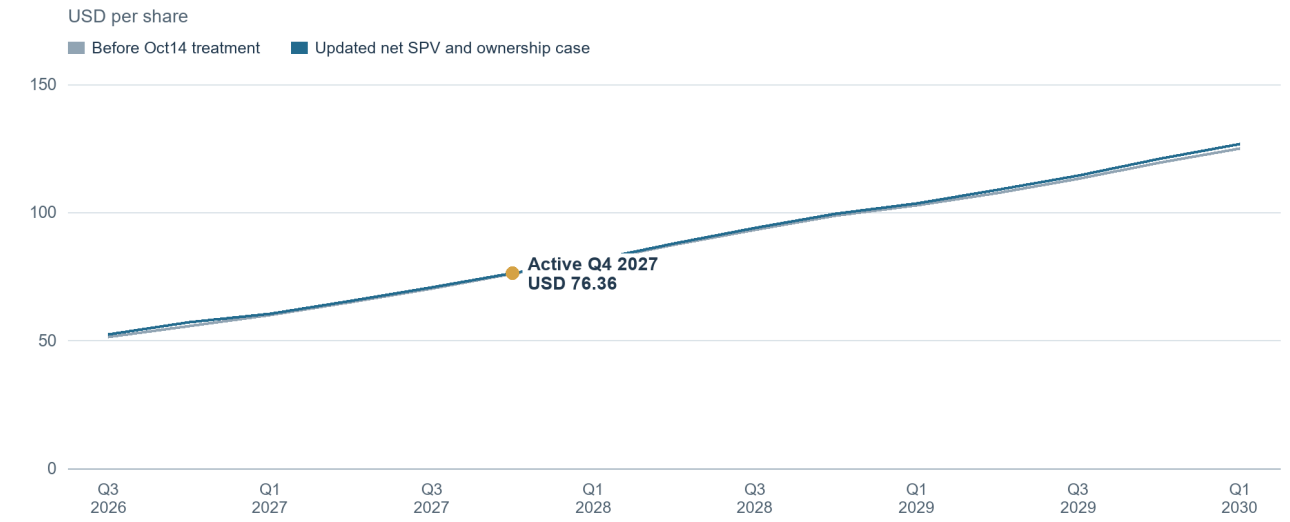
Margins against existing revenue reference



Denominator excludes unallocated Japan fees; OP/reference revenue is a proxy, not reported consolidated operating margin.

Quarter	Product GP / reference	OP / reference proxy
Q3 2026	30.500%	16.707%
Q4 2026	32.144%	17.969%
Q1 2027	33.041%	19.357%
Q2 2027	34.015%	20.404%
Q3 2027	34.993%	21.365%
Q4 2027	35.932%	22.402%
Q1 2028	37.227%	23.235%
Q2 2028	37.780%	24.029%
Q3 2028	38.296%	24.805%
Q4 2028	38.858%	25.565%
Q1 2029	40.481%	27.017%
Q2 2029	40.669%	27.350%
Q3 2029	40.888%	27.668%
Q4 2029	41.071%	27.977%
Q1 2030	42.417%	29.318%
Q2 2030	42.572%	29.750%
Q3 2030	42.725%	30.201%
Q4 2030	42.908%	30.695%

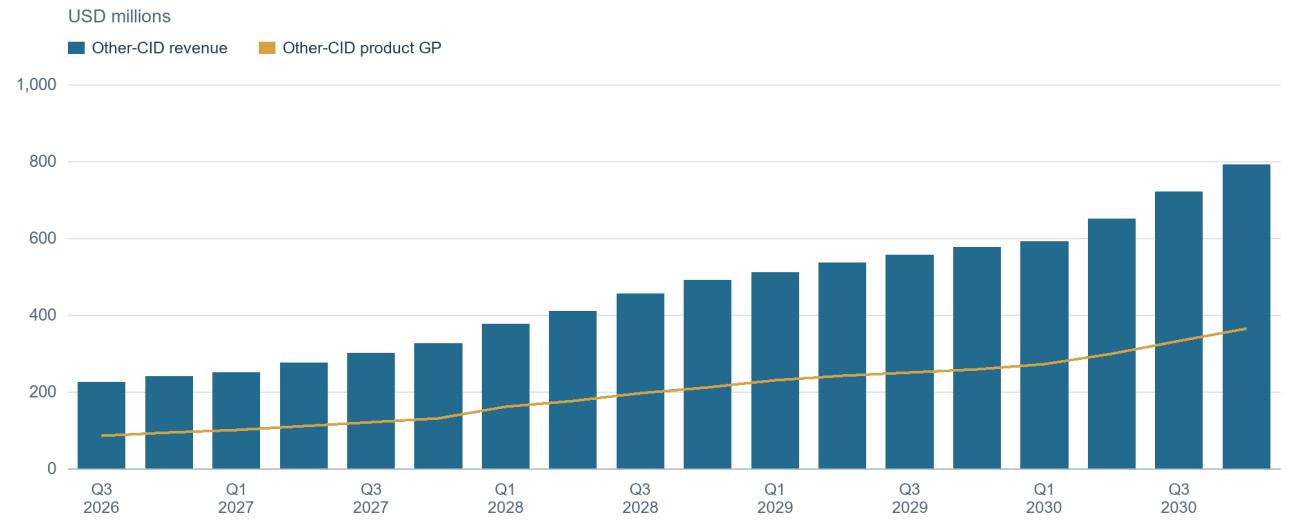
Rolling four-quarter modeled ceiling



Retained 20x forward non-IFRS OP and target-quarter base shares; separate expected ownership factor 0.976 from 2027. Issuance is conditional.

Quarter	Updated ceiling USD
Q3 2026	52.344964
Q4 2026	57.168447
Q1 2027	60.310244
Q2 2027	65.357831
Q3 2027	70.673314
Q4 2027	76.360690
Q1 2028	81.393718
Q2 2028	87.771117
Q3 2028	93.866376
Q4 2028	99.452295
Q1 2029	103.589647
Q2 2029	108.737665
Q3 2029	114.399846
Q4 2029	120.827695
Q1 2030	126.781979

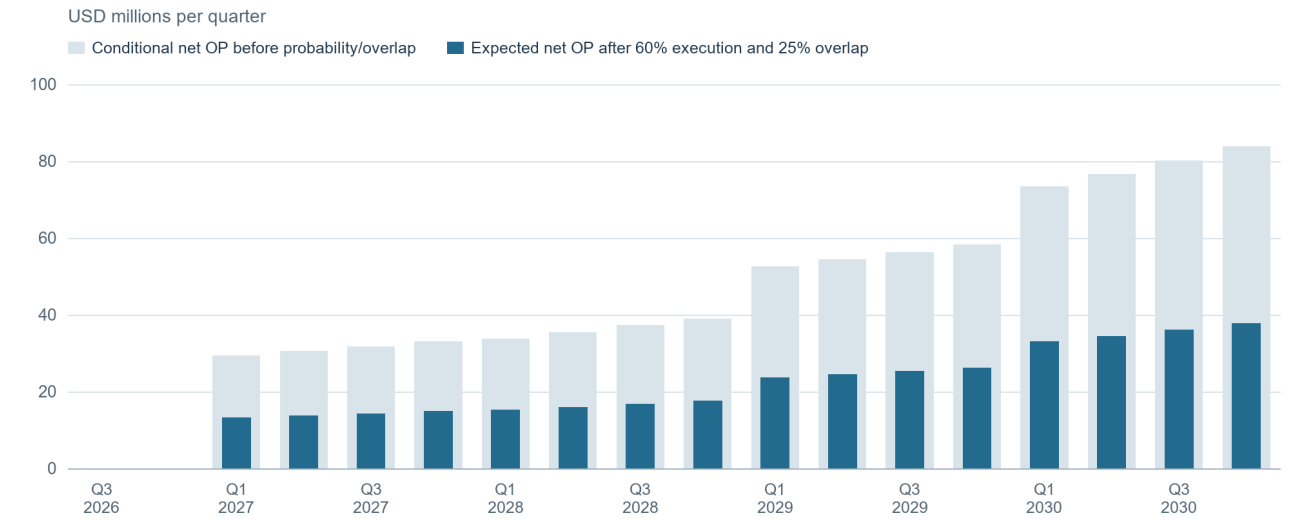
Other CID: retained underlying ramp



Existing SiGe, power and communications forecast retained. No SPV wafer sales, grant windfall or repeat contract is added.

Quarter	Other-CID revenue	Product GP
Q3 2026	225.000	85.500
Q4 2026	240.000	93.600
Q1 2027	250.000	100.000
Q2 2027	275.000	110.000
Q3 2027	300.000	120.000
Q4 2027	325.000	130.000
Q1 2028	375.000	161.250
Q2 2028	410.000	176.300
Q3 2028	455.000	195.650
Q4 2028	490.000	210.700
Q1 2029	510.000	229.500
Q2 2029	535.000	240.750
Q3 2029	555.000	249.750
Q4 2029	575.000	258.750
Q1 2030	590.000	271.400
Q2 2030	650.000	299.000
Q3 2030	720.000	331.200
Q4 2030	790.000	363.400

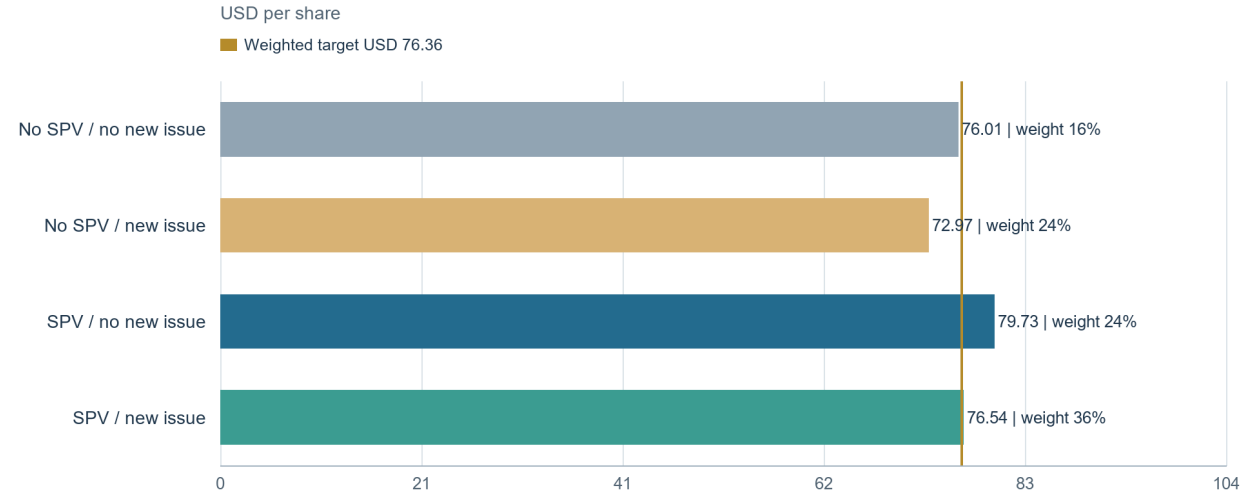
Japan SPV: bounded net operating contribution



Analyst net-OP proxy: reference revenue x 1.5% in 2027/28, 2.0% in 2029 and 2.5% in 2030 x 60% x 75%; gross fees remain unallocated.

Quarter	Reference revenue	Conditional net OP	Expected net OP
Q3 2026	1,885.000	0.000	0.000
Q4 2026	1,940.000	0.000	0.000
Q1 2027	1,950.000	29.250	13.163
Q2 2027	2,030.000	30.450	13.702
Q3 2027	2,115.000	31.725	14.276
Q4 2027	2,200.000	33.000	14.850
Q1 2028	2,250.000	33.750	15.188
Q2 2028	2,360.625	35.409	15.934
Q3 2028	2,481.875	37.228	16.753
Q4 2028	2,592.500	38.887	17.499
Q1 2029	2,621.875	52.438	23.597
Q2 2029	2,722.500	54.450	24.503
Q3 2029	2,812.500	56.250	25.312
Q4 2029	2,913.125	58.263	26.218
Q1 2030	2,932.500	73.312	32.991
Q2 2030	3,062.500	76.562	34.453
Q3 2030	3,202.500	80.062	36.028
Q4 2030	3,352.500	83.812	37.716

Q4 2027: conditional execution and ownership states



Independent 60% execution and 60% new-issuance judgments produce four conditional states. Actual transfer form and terms are unknown.

SPV executes	New issue	Weight	F4Q OP	State shares M	State target
No	No	16%	2,135.775000	562.000000	76.006228
No	Yes	24%	2,135.775000	585.416667	72.965979
Yes	No	24%	2,240.315625	562.000000	79.726535
Yes	Yes	36%	2,240.315625	585.416667	76.537473

6. Revenue sources, capture and product margins

Calendar quarters; preserve 20x non-IFRS OP, not EBITDA. Separate SiPho, other CID (including SiGe and data-center power), and all other markets/services. These mutually exclusive buckets avoid adding IP/services a second time. Q3 product revenue is an analyst split of the consolidated midpoint, not disclosed segment guidance. Its product GMs reconcile exactly to 30.5% consolidated GM and USD 260M opex. Subsequent quarterly revenue, mix and margins are LOW-confidence price target assumptions. SiPho reaches USD 1.08B annualized in Q4 2028 (actual model FY2028 sales USD 900M) and USD 2.16B annualized in Q4 2030 (FY2030 USD 1.95B). Other CID includes the existing Marvell relationship and MPS-related data-center power; the rest of MPS end-market exposure remains in other markets. No additional agreement-value layer is added. Legacy margins recover with utilization and mix, while R&D/opex rises in dollars. Displayed product GMs are analytical allocations, not company disclosures. Shares start at guided 556M, then rise to 562/568/574/580M for 2027-30 to allow for continuing equity compensation; no unexecuted buyback benefit is assumed. Grants are not added to OP or revenue. Full future option dilution is not independently reconstructed. The new SPV branch is separate net OP with unallocated gross fees, and the government ownership risk is separate from this underlying share schedule.

Vector	Actual calibration / source	Forward demand and capture	Margin derivation / constraint
Silicon photonics	85M Q3 allocation; company discloses a broader CID segment.	FY27 sales 585M, FY28 900M, FY29 1,425M, FY30 1,950M. FY30/FY27=3.333333×. An illustrative 2.5×productive wafer-equivalent output ×1.333333 die/process content provides the bridge; neither factor is disclosed capacity guidance. The module-to-wafer capture is not inferred from total optical TAM alone.	52% 2027→57% 2029-30 non-IFRS product GM proxy. Implied product cost falls 48→43% with loading/yield and differentiated mix; manufacturing depreciation remains included.
Other CID	225M Q3 allocation includes SiGe, power and communications.	Annual 1,150/1,730/2,175/2,750M. Marvell capacity and MPS power are within these totals. At an illustrative 3,000 USD per recognized wafer-equivalent invoice,2030 requiresabout 916,667 equivalents; this is a sensitivity denominator, not a disclosed shipment forecast.	40% 2027→46% 2030 proxy. Increased content may also raise mask/process cost, so revenue growth alone cannot establish the margin.
Other markets/ services	1,575M Q3 residual.	Annual 6,560/6,970/7,300/7,680M. Growth reflects a gradual recovery and richer automotive/IoT/IP mix, not optical growth applied to the whole company. Technology services are already included.	2027 GM 30.5→33.5%,2028 34→35.5%,2029 36.5%,2030 38%. These are demanding mix/productivity assumptions, not current broad-market guidance.

Confidence: reported historical totals are HIGH. Source-based forward extrapolations are MEDIUM only where the cited source sizes the relevant period; other product allocations, future capture and margin assumptions are LOW. A source for the historical anchor does not make an unsourced future factor management guidance. The detailed factors below are the single modeled base case, with uncertain assumptions exposed for review.

V1. Silicon photonics (within CID)

Sources: G1/2/3/7; product allocation and exit-run-rate trajectory. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2026	85.000000	1.000000	85.000	50.000%	42.500
Q4 2026	85.000000	1.176471	100.000	50.000%	50.000
Q1 2027	85.000000	1.411765	120.000	52.000%	62.400
Q2 2027	85.000000	1.588235	135.000	52.000%	70.200
Q3 2027	85.000000	1.823529	155.000	52.000%	80.600
Q4 2027	85.000000	2.058824	175.000	52.000%	91.000
Q1 2028	85.000000	2.176471	185.000	55.000%	101.750
Q2 2028	85.000000	2.470588	210.000	55.000%	115.500
Q3 2028	85.000000	2.764706	235.000	55.000%	129.250
Q4 2028	85.000000	3.176471	270.000	55.000%	148.500
Q1 2029	85.000000	3.529412	300.000	57.000%	171.000
Q2 2029	85.000000	3.941176	335.000	57.000%	190.950
Q3 2029	85.000000	4.411765	375.000	57.000%	213.750
Q4 2029	85.000000	4.882353	415.000	57.000%	236.550
Q1 2030	85.000000	5.176471	440.000	57.000%	250.800
Q2 2030	85.000000	5.529412	470.000	57.000%	267.900

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2030	85.000000	5.882353	500.000	57.000%	285.000
Q4 2030	85.000000	6.352941	540.000	57.000%	307.800

V2. Other CID: SiGe, power and communications

Sources: G2/4/5; SiGe and power demand, undisclosed contract values. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2026	225.000000	1.000000	225.000	38.000%	85.500
Q4 2026	225.000000	1.066667	240.000	39.000%	93.600
Q1 2027	225.000000	1.111111	250.000	40.000%	100.000
Q2 2027	225.000000	1.222222	275.000	40.000%	110.000
Q3 2027	225.000000	1.333333	300.000	40.000%	120.000
Q4 2027	225.000000	1.444444	325.000	40.000%	130.000
Q1 2028	225.000000	1.666667	375.000	43.000%	161.250
Q2 2028	225.000000	1.822222	410.000	43.000%	176.300
Q3 2028	225.000000	2.022222	455.000	43.000%	195.650
Q4 2028	225.000000	2.177778	490.000	43.000%	210.700
Q1 2029	225.000000	2.266667	510.000	45.000%	229.500
Q2 2029	225.000000	2.377778	535.000	45.000%	240.750
Q3 2029	225.000000	2.466667	555.000	45.000%	249.750
Q4 2029	225.000000	2.555556	575.000	45.000%	258.750
Q1 2030	225.000000	2.622222	590.000	46.000%	271.400
Q2 2030	225.000000	2.888889	650.000	46.000%	299.000
Q3 2030	225.000000	3.200000	720.000	46.000%	331.200
Q4 2030	225.000000	3.511111	790.000	46.000%	363.400

V3. Other markets and services, net of CID

Sources: G1/2/7; remaining end markets including technology services. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2026	1,575.000000	1.000000	1,575.000	28.376%	446.925
Q4 2026	1,575.000000	1.015873	1,600.000	30.000%	480.000
Q1 2027	1,575.000000	1.003175	1,580.000	30.500%	481.900
Q2 2027	1,575.000000	1.028571	1,620.000	31.500%	510.300

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2027	1,575.000000	1.053968	1,660.000	32.500%	539.500
Q4 2027	1,575.000000	1.079365	1,700.000	33.500%	569.500
Q1 2028	1,575.000000	1.073016	1,690.000	34.000%	574.600
Q2 2028	1,575.000000	1.098413	1,730.000	34.500%	596.850
Q3 2028	1,575.000000	1.117460	1,760.000	35.000%	616.000
Q4 2028	1,575.000000	1.136508	1,790.000	35.500%	635.450
Q1 2029	1,575.000000	1.130159	1,780.000	36.500%	649.700
Q2 2029	1,575.000000	1.149206	1,810.000	36.500%	660.650
Q3 2029	1,575.000000	1.168254	1,840.000	36.500%	671.600
Q4 2029	1,575.000000	1.187302	1,870.000	36.500%	682.550
Q1 2030	1,575.000000	1.180952	1,860.000	38.000%	706.800
Q2 2030	1,575.000000	1.206349	1,900.000	38.000%	722.000
Q3 2030	1,575.000000	1.231746	1,940.000	38.000%	737.200
Q4 2030	1,575.000000	1.257143	1,980.000	38.000%	752.400

7. Operating-cost bridge and quarterly operating model

Existing expense dollars begin at USD 260 million in Q3 2026 and USD 275 million in Q4; R&D and selling costs rise through the model. IP/design-team acquisitions require spending before wafer revenue matures. Product GP includes manufacturing depreciation; no EBITDA add-back enters valuation. The retained non-IFRS basis excludes recurring SBC. Q3 guidance assumes approximately USD 76 million SBC; a share allowance does not erase that economic compensation cost.

Japan-SPV net OP is a separate service/technology contribution after associated program costs. It enters neither product GP nor an invented gross-fee revenue line. Updated OP = product GP - existing opex + expected net SPV contribution. OP/reference revenue is a proxy until gross fee presentation is disclosed.

Quarter	Revenue ref.	Product GP	Existing opex	Net SPV OP	Updated OP	OP / ref. proxy
Q3 2026	1,885.000	574.925	260.000	0.000	314.925	16.707%
Q4 2026	1,940.000	623.600	275.000	0.000	348.600	17.969%
Q1 2027	1,950.000	644.300	280.000	13.163	377.462	19.357%
Q2 2027	2,030.000	690.500	290.000	13.702	414.203	20.404%
Q3 2027	2,115.000	740.100	302.500	14.276	451.876	21.365%
Q4 2027	2,200.000	790.500	312.500	14.850	492.850	22.402%
Q1 2028	2,250.000	837.600	330.000	15.188	522.788	23.235%
Q2 2028	2,360.625	891.838	340.531	15.934	567.240	24.029%
Q3 2028	2,481.875	950.462	351.594	16.753	615.621	24.805%
Q4 2028	2,592.500	1,007.400	362.125	17.499	662.774	25.565%
Q1 2029	2,621.875	1,061.356	376.594	23.597	708.359	27.017%
Q2 2029	2,722.500	1,107.225	387.125	24.503	744.602	27.350%

Quarter	Revenue ref.	Product GP	Existing opex	Net SPV OP	Updated OP	OP / ref. proxy
Q3 2029	2,812.500	1,149.975	397.125	25.312	778.162	27.668%
Q4 2029	2,913.125	1,196.444	407.656	26.218	815.006	27.977%
Q1 2030	2,932.500	1,243.875	417.125	32.991	859.741	29.318%
Q2 2030	3,062.500	1,303.775	427.125	34.453	911.103	29.750%
Q3 2030	3,202.500	1,368.275	437.125	36.028	967.178	30.201%
Q4 2030	3,352.500	1,438.475	447.125	37.716	1,029.066	30.695%

Every row remains Model, including the Q3 2026 guidance midpoint. Displayed rounding is presentation only. Product GP and existing opex are unchanged; the net contribution is added once.

8. Multiple, shares and recognition convention

The locked method remains 20x forward-four-quarter consolidated non-IFRS operating profit, including manufacturing depreciation and excluding recurring SBC. It is neither EBITDA, after-tax EPS nor an enterprise-value calculation. No framework-wide multiple replaces this company-specific method. Existing product GP less existing opex supplies core OP; the net SPV contribution is added separately.

Underlying target-quarter shares remain 556 million in Q3 2026, 557 million in Q4 2026 and 562/568/574/580 million in 2027-30. Annual additions are a net award/acquisition allowance, not a reported future share forecast. No unexecuted repurchase benefit is assumed. Historical Q2 cash/equivalents/marketable securities were about USD 3.3 billion versus debt USD 1.1 billion; these are not an October 14 balance-sheet refresh. Customer prepayments and grants carry obligations. Cash, debt and exercise proceeds are not separately capitalized.

From target-quarter 2027, a separate conditional ownership state assumes 60% probability of 4% post-issue new common equity and 40% transfer/no-issue probability. Expected per-share factor 0.976 equals the exact weighted state effect, not an asserted issued share count or replacement of base shares. Before 2027 the factor is 1 under the retained target-quarter convention. Actual form, consideration, date and shares remain unknown. Execution and issuance are assumed independent, subject to revision when contracts arrive.

Unit factor is 1, yielding USD per share. No cash/debt bridge, tax substitution, multiple change or completed-issuance claim is introduced.

9. Complete rolling four-quarter price-target derivation

Expected OP includes the separate net SPV contribution. The ownership factor is 1 in 2026 and 0.976 from 2027. This calculation equals the four-state weighted target under the disclosed independence assumption. Base shares remain visible. Amounts are USD millions except per-share values.

Target quarter	Earnings window	Forward OP	Multiple	Base shares M	Factor	Ceiling USD
Q3 2026	Q3 2026 to Q2 2027	1,455.190000	20x	556.000	1.000	52.344964
Q4 2026	Q4 2026 to Q3 2027	1,592.141250	20x	557.000	1.000	57.168447
Q1 2027	Q1 2027 to Q4 2027	1,736.391250	20x	562.000	0.976	60.310244
Q2 2027	Q2 2027 to Q1 2028	1,881.716250	20x	562.000	0.976	65.357831
Q3 2027	Q3 2027 to Q2 2028	2,034.754219	20x	562.000	0.976	70.673314
Q4 2027	Q4 2027 to Q3 2028	2,198.499375	20x	562.000	0.976	76.360690
Q1 2028	Q1 2028 to Q4 2028	2,368.423750	20x	568.000	0.976	81.393718
Q2 2028	Q2 2028 to Q1 2029	2,553.995625	20x	568.000	0.976	87.771117
Q3 2028	Q3 2028 to Q2 2029	2,731.357656	20x	568.000	0.976	93.866376

Target quarter	Earnings window	Forward OP	Multiple	Base shares M	Factor	Ceiling USD
Q4 2028	Q4 2028 to Q3 2029	2,893.898750	20x	568.000	0.976	99.452295
Q1 2029	Q1 2029 to Q4 2029	3,046.130000	20x	574.000	0.976	103.589647
Q2 2029	Q2 2029 to Q1 2030	3,197.511250	20x	574.000	0.976	108.737665
Q3 2029	Q3 2029 to Q2 2030	3,364.011875	20x	574.000	0.976	114.399846
Q4 2029	Q4 2029 to Q3 2030	3,553.027500	20x	574.000	0.976	120.827695
Q1 2030	Q1 2030 to Q4 2030	3,767.087500	20x	580.000	0.976	126.781979

- **Q3 2026:** (314.925000 + 348.600000 + 377.462500 + 414.202500) x 20 / 556.000000 x 1.000000 = **52.344964 USD/share**. Window: Q3 2026 to Q2 2027.

- **Q4 2026:** (348.600000 + 377.462500 + 414.202500 + 451.876250) x 20 / 557.000000 x 1.000000 = **57.168447 USD/share**. Window: Q4 2026 to Q3 2027.

- **Q1 2027:** (377.462500 + 414.202500 + 451.876250 + 492.850000) x 20 / 562.000000 x 0.976000 = **60.310244 USD/share**. Window: Q1 2027 to Q4 2027.

- **Q2 2027:** (414.202500 + 451.876250 + 492.850000 + 522.787500) x 20 / 562.000000 x 0.976000 = **65.357831 USD/share**. Window: Q2 2027 to Q1 2028.

- **Q3 2027:** (451.876250 + 492.850000 + 522.787500 + 567.240469) x 20 / 562.000000 x 0.976000 = **70.673314 USD/share**. Window: Q3 2027 to Q2 2028.

- **Q4 2027:** (492.850000 + 522.787500 + 567.240469 + 615.621406) x 20 / 562.000000 x 0.976000 = **76.360690 USD/share**. Window: Q4 2027 to Q3 2028.

- **Q1 2028:** (522.787500 + 567.240469 + 615.621406 + 662.774375) x 20 / 568.000000 x 0.976000 = **81.393718 USD/share**. Window: Q1 2028 to Q4 2028.

- **Q2 2028:** (567.240469 + 615.621406 + 662.774375 + 708.359375) x 20 / 568.000000 x 0.976000 = **87.771117 USD/share**. Window: Q2 2028 to Q1 2029.

- **Q3 2028:** (615.621406 + 662.774375 + 708.359375 + 744.602500) x 20 / 568.000000 x 0.976000 = **93.866376 USD/share**. Window: Q3 2028 to Q2 2029.

- **Q4 2028:** (662.774375 + 708.359375 + 744.602500 + 778.162500) x 20 / 568.000000 x 0.976000 = **99.452295 USD/share**. Window: Q4 2028 to Q3 2029.

- **Q1 2029:** (708.359375 + 744.602500 + 778.162500 + 815.005625) x 20 / 574.000000 x 0.976000 = **103.589647 USD/share**. Window: Q1 2029 to Q4 2029.

- **Q2 2029:** (744.602500 + 778.162500 + 815.005625 + 859.740625) x 20 / 574.000000 x 0.976000 = **108.737665 USD/share**. Window: Q2 2029 to Q1 2030.

- **Q3 2029:** (778.162500 + 815.005625 + 859.740625 + 911.103125) x 20 / 574.000000 x 0.976000 = **114.399846 USD/share**. Window: Q3 2029 to Q2 2030.

- **Q4 2029:** (815.005625 + 859.740625 + 911.103125 + 967.178125) x 20 / 574.000000 x 0.976000 = **120.827695 USD/share**. Window: Q4 2029 to Q3 2030.

- **Q1 2030:** (859.740625 + 911.103125 + 967.178125 + 1029.065625) x 20 / 580.000000 x 0.976000 = **126.781979 USD/share**. Window: Q1 2030 to Q4 2030.

10. Annual trajectory cross-check

Calendar-year USD millions. Revenue reference and product GP exclude unallocated SPV gross fees. Existing opex stays on its original path; program costs are already netted in the new contribution. OP/reference is a proxy rather than reported consolidated margin. Annual sums add no second layer of economics.

Year	Revenue ref.	Product GP	Existing opex	Net SPV OP	Updated OP	OP / ref. proxy
2027	8,295.000	2,865.400	1,185.000	55.991	1,736.391	20.933%
2028	9,685.000	3,687.300	1,384.250	65.374	2,368.424	24.455%
2029	11,070.000	4,515.000	1,568.500	99.630	3,046.130	27.517%
2030	12,550.000	5,354.400	1,728.500	141.188	3,767.088	30.017%

Original revenue reference remains USD 8,295/9,685/11,070/12,550 million in 2027-30, including the existing interposer branch once. These calendar sums are not the live valuation window. SiPho full-year 2028/2030 sales remain USD 900/1,950 million, whereas Q4 exit sales USD 270/540 million imply annualized USD 1.08/2.16 billion. An exit rate cannot be placed in every quarter without accelerating recognition.

Issuer EPS accretion is only an after-tax cross-check. It is not inserted into OP or capitalized at the OP multiple. Phase-two economics remain zero pending approval and a defensible timing/cost bridge.

11. Historical Q3 2027 dilution diagnostic

This table is the September 17 snapshot and retains its stated Q3 2027 window and pre-interposer operating case. The current active Q4 2027 target is $76.360690; the older diagnostic is not an economic downgrade or an alternative active horizon.

Denominator sensitivity	Q3 2027 shares M	Q3 2027 price target USD	Effect on price target
-5% shares	533.900000	74.053193	+5.26%
+0% shares	562.000000	70.350534	+0.00%
+5% shares	590.100000	67.000508	-4.76%
+10% shares	618.200000	63.955031	-9.09%

This isolates denominator uncertainty; it is not a second bull/bear operating case or permission to remove known dilution. The current base case remains the explicit share path in the rolling table; the historical diagnostic is not recalculated as a new case.

12. Programme and probability ledger

Programme / signal	Source and probability	Amount / unit base	Treatment in this model	Escalation / recognition
Existing optical production	Reported end-market growth;95% continuation confidence	CID actuals and qualified processes	Calibrates the initial allocation	Actual SiPho/SiGe disclosure would replace analytical splits.
Marvell SiGe agreement	Named customer partnership;85% execution confidence	Undisclosed volumes/pricing	Inside other CID; no added agreement value	Qualified Vermont capacity and customer shipments.
MPS production	Signed partnership;80% timing confidence	Early 2027 start; undisclosed revenue	Data-center portion inside CID, remaining applications in other markets	Process transfer, yield and volume start.
SCALE NPO/CPO	Customer tape-outs;70% conversion confidence	No complete public unit schedule	Inside SiPho trajectory, not another whole market	Customer qualification and accepted volume sales.
TSMC interposer supply	Signed agreement; 85% execution scenario	$2B headline, initial five-year term; H1 2028 expected ramp	Explicit accepted-sales schedule with 50% overlap and full incremental cost bridge	Malta tooling, qualified output, acceptance, capex/payment terms and realized margin.
Japan-funded phase-one services	October 14 issuer release/deck; 60% execution judgment	Conditional net OP: 1.5% of reference revenue in 2027/28, 2.0% in 2029, 2.5% in 2030	25% overlap reserve; costs already netted; gross fees unallocated; no wafer sales or funding windfall	Definitive agreements, accounting, cost allocation, closing and fee recognition; phase two zero.
Commerce equity agreement	Approximately 4% announced; 60% new-issue / 40% transfer/no-issue judgments	Conditional 4% post-issue stake; actual terms undisclosed	Expected factor 0.976 from target-quarter 2027, separate from base shares and execution	Verify issue versus transfer, exact shares, consideration, date and dependence on closing.
Quantum support	Finalized milestone award;85% funding confidence	Up to 375M overfiveyears	No immediate sales/OP windfall	Grant accounting and milestones must be reconciled separately.
Photonics funding	Letter of intent;60% realization confidence	300M proposal	No modeled OP uplift	Definitive agreement and recognized accounting treatment.

Probabilities are analyst judgments under the framework signal hierarchy. Embedded revenue is not multiplied or added a second time. Where no defensible amount or unit base exists, the opportunity remains explicitly unquantified; confidence alone cannot create a revenue number. New evidence can enlarge the TAM, improve capture or move recognition earlier, but each requires an identifiable model input.

13. Scored risk classes

Risk class	Scored deduction	Mechanism and consequence
Capex and qualification	-1	Tools, yield and customer qualification determine usable capacity and cash returns.
Customer concentration	-1	A few large customers and programme changes can move whole technology corridors.
Competition and legacy cycle	-1	Mobile/auto softness and competing foundries can offset optical growth.
Adjusted economics and dilution	-1	Recurring SBC is excluded; acquisitions and future awards dilute common claims. Government equity adds conditional new-issuance risk with unknown final terms.
Policy and supply chain	-1	Geopolitics, funding milestones and critical materials can change cost, timing and demand.

14. Dated catalysts, counters and escalation

1. **2026-11-04, 08:30 EST - scheduled Q3 results call [G12].** Test 1,885M revenue,30.5% GM and 260M expenses.

2. **2027-03-31 - MPS initial production** Confirm process qualification and recognized shipments; a partnership announcement is insufficient.

3. **2027-12-31 - SiGe and NPO** Track usable capacity, pricing and converted tape-outs.

4. **2028-12-31 - SiPho exit milestone** Compare actual quarter with 270M model; distinguish annual revenue from annualized run rate.

5. **2030-12-31 - Larger photonics case** 540M exit quarter requires funded, qualified output and customer capture.

6. **Every report - Costs and shares** Check manufacturing cost, R&D, SBC and net dilution against the path.

7. **Next signed agreements / 2027 fee start:** Verify SPV close, gross fee recognition, program/shared cost allocation and government share-transfer form. Reconcile execution, overlap and issuance assumptions to new evidence; illustrations are not guarantees.

15. Cross-book monitoring

Marvell is a verified SiGe customer through the September 17 capacity agreement. IQE/Tower and GF offer competing or complementary technology routes, not assured simultaneous capture of every optical link. Foundry revenue and downstream component sales represent different invoices within the same end-demand pool. No unverified exclusive AMD device attribution or fixed Sivers-to-GF capacity multiplier is used. MRVL's use of GF SiGe does not alone establish a loss at Tower: product process, customer socket and sourcing share must be identified. IQE epi content and external laser suppliers may serve platform-specific designs without growing in a universal fixed ratio to GF revenue.

16. Source register and model limits

G9 - Japan-funded Malta two-fab initiative (2026-10-14). Primary issuer release, fully recovered through its official German version and issuer-distributed English release. Funding, operator role, 2029 timing, capacity, award reallocation and approximately 4% equity are disclosed. Exact share mechanics are not.

G10 - Advancing Semiconductor Leadership in America (2026-10-14). Official 16-page deck, full supported web text. Pages 8-12 identify approval conditions, government-owned SPV, funding/cost roles, 2027 fees and phase-one-only non-IFRS illustrations. Original native PDF reads returned HTTP403; screenshot tools supplied placeholders rather than pixels. No original-slide visual review is claimed. Contracts/accounting remain conditions.

G11 - October 14 SEC Form 6-K and Exhibit 99.1. Primary filing contains the same press release, with no separate equity/SPV contract. It confirms neither issuance versus holder transfer, price, pre/post basis, exact shares, vesting nor timing.

G12 - Issuer investor-relations event schedule (retrieved 2026-10-14). Q3 2026 results call is scheduled November 4, 2026 at 08:30 EST. This replaces an older November 15 monitoring date; it is not a new operating input.

Supplemental access limit: The official webcast landing page supplied no readable transcript. A nonissuer prepared-remarks reproduction attributes about USD 50 million 2027 revenue and construction-fee presentation comments to the CFO, but has obvious transcription errors and was not checked against primary audio. These additional amounts are not used as issuer-verified inputs; no full issuer Q&A review is claimed.

G8 - TSMC interposer manufacturing agreement (2026-10-08). Primary issuer announcement. Agreement existence is verified; timing, acceptance, payment mix and realized margin remain uncertain. The October 11 Grok relay and the original announcement are one economic event.

G1 - Q2 results and Q3 guidance (2026-08-05). Reported financials and Q3 midpoint anchor the first forecast quarter. SBC remains excluded from the chosen non-IFRS basis.

G2 - Q2 management call (2026-08-05). CID was 16% of Q2 revenue, +62% year on year. SiPho is expected to more than double in 2026; SiGe capacity is oversubscribed through 2027. NPO/CPO ramps were discussed for 2027/2028.

Technology-services revenue is already inside end-market totals.

G3 - Goldman Sachs management discussion (2026-09-08). The discussion refers to SiPho USD 1B exit-2028 and USD 2B 2030 run-rate milestones. Capacity build and qualification constrain near-term conversion. These are not full-year revenue figures.

G4 - Expanded Marvell SiGe agreement (2026-09-17). Multi-year agreement expands Vermont SiGe capacity for Marvell optical requirements. No contract revenue, wafer volume or price is disclosed.

G5 - MPS manufacturing partnership (2026-09-09). MPS proprietary processes will be deployed in Singapore, with volume production expected in early 2027. No revenue amount is stated.

G6 - Quantum funding agreement (2026-09-08). Up to USD 375M over five years, conditional on milestones. The separate USD 300M photonics program remains an LOI in this release; neither is booked as an immediate earnings windfall.

G7 - Prior-quarter management transcript (2026-05-05). Cross-check confirms the prior SiPho exit-run-rate framing and SiGe capacity constraints.

Product-level revenue and margins, outer-year growth and share trajectories are assumptions. No complete two-call Q&A review. The framework values consolidated adjusted OP without a cash/debt bridge, taxes or SBC expense; it is a conditional target rather than fair value. Added net SPV OP is after program costs; gross fees remain unallocated. A conditional ownership state does not establish completed issuance.

Workbook: v3_2_master_portfolio.xlsx, model tab GFS_Q_Model. Portfolio ranking, prices and return calculations stay in the master workbook. **October 11 update ID:** GROK-20261011-2105-GFS-INTERPOSERS. Grok XDATA-20261011-2003-ET-BQYV R0, source posts 2109355351099556085 and 2109427215410852327/2109427277125853186, reconciled to the October 8 issuer event.

Current artifact: G:/My Drive/1amInvesting/Artifacts/GFS_GlobalFoundries_DeepDive_Q2_2026.pdf, with synchronized self-contained Markdown. **Original snapshot:** versions/originals/GFS_GlobalFoundries_DeepDive_Q2_2026.md. Superseded history is retained separately in VERSION_LOG, not in this current-state model.

October 14 update ID: DOTX-20261014-1958-GFS-JAPAN. Net phase-one service economics and conditional Commerce-equity risk are modeled; publication/recovery status is recorded separately in the ledger.